MATH73235 Spring 2025 Quiz 1

First name: Hayden

Last name: Foster

Student ID:

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

$\frac{d}{dt}\cos = -$

5

[5 marks]

Solution.

$3\frac{d}{dt}\cos(2t)$

$= -6\sin(2t)$

$3\frac{d}{dt}\sin(2t)$

$= 6\cos(2t)$

$4\frac{d}{dt}t$

$= 4$

$\vec{r}\,' = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix}$ ✓ $\quad \vec{r}\,'' = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix}$ ✓

$-6\frac{d}{dt}\sin(2t)$

$= -12\cos(2t)$

$6\frac{d}{dt}\cos(2t)$

$= -12\sin(2t)$

$\frac{d}{dt}4$

$= 0$

□

MATH73235 Spring 2025 Quiz 1

First name: Zach

Last name: McEwen

Student ID: 8768637

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$\vec{r}'(t) = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix}$$

$$r''(t) = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix}$$

$\square$

MATH73235 | Spring 2025 | Quiz 1
First name: Jesse ~~Russell~~
Last name: Russell
Student ID: 7662240

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$\vec{r}\,'(t) = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix} \checkmark$$

$$\vec{r}\,''(t) = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix} \checkmark$$

□

First name: Kyle
Last name: Dick
Student ID: 7206287

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

[5 marks]

Solution.

$$\vec{r}\,'(t) = \begin{bmatrix} -6 \sin(2t) \\ 6 \cos(2t) \\ 4 \end{bmatrix}$$

$$\vec{r}\,''(t) = \begin{bmatrix} -12 \cos(2t) \\ -12 \sin(2t) \\ 0 \end{bmatrix}$$

□

MATH73235 Spring 2025 Quiz 1

First name: JOSHUA

Last name: DIPLOMA

Student ID: 7897960

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$r'(t) = \frac{\partial}{\partial t}\begin{bmatrix} 3\cos 2t \\ 3\sin 2t \\ 4t \end{bmatrix} \qquad r'' = \frac{\partial}{\partial t}\begin{bmatrix} -6\sin 2t \\ 6\cos 2t \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} (2)(-3\sin 2t) \\ (2)(3\cos 2t) \\ 4 \end{bmatrix} \qquad r'' = \begin{bmatrix} (2)(-6\cos 2t) \\ (2)(-6\sin 2t) \\ 0 \end{bmatrix}$$

$$\boxed{r'(t) = \begin{bmatrix} -6\sin 2t \\ 6\cos 2t \\ 4 \end{bmatrix} \quad r'' = \begin{bmatrix} -12\cos 2t \\ -12\sin 2t \\ 0 \end{bmatrix}}$$

□

MATH73235 Spring 2025 Quiz 1

First name: Coulter

Last name: Rolph

Student ID: 875 1065

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$\frac{d}{dt} = \cos(t)$

$= -\sin(t)$

$\frac{d}{dt} = \sin(t)$

$= \cos(t)$

$$\vec{r}'(t) = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix}$$ ✓

$$\vec{r}''(t) = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix}$$ ✓

□

First name:
Last name:
Student ID:

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$r'(t) = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix} \checkmark$$

$$r''(t) = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix} \checkmark$$

$\square$

MATH73235 Spring 2025 Quiz 1

First name: AMY

Last name: WENTZELL

Student ID: 8820998

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$\vec{r}\,'(t) = \begin{bmatrix} -3\sin(2t)(2) \\ 3\cos(2t)\;(2) \\ 4 \end{bmatrix} = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix}$$ ✓

$$\vec{r}\,''(t) = \begin{bmatrix} -(2)6\cos(2t) \\ -(2)6\sin(2t) \\ 0 \end{bmatrix} = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix}$$ ✓

$\therefore\ \vec{r}\,'(t) = [-6\sin(2t),\ 6\cos(2t),\ 4]$

$\therefore\ \vec{r}\,''(t) = [-12\cos(2t),\ -12\sin(2t),\ 0]$

□

MATH73235 Spring 2025 Quiz 1

First name: Stuart

Last name: Murray

Student ID: 3864228

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix} \longrightarrow \vec{r}\,'(t) = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix} \longrightarrow \vec{r}\,''(t) = \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ \emptyset \end{bmatrix}$$

$$\frac{d}{dt}\cos(t) = -\sin(t)$$

$$\frac{d}{dt}\sin(t) = \cos(t)$$

□

MATH73235 Spring 2025 Quiz 1

First name: Alan
Last name: Hosseinpour moghadam
Student ID: 8825132

Problem 2. *Find the first and second derivatives (with respect to t) of*

$$\vec{r}(t) = \begin{bmatrix} 3\cos(2t) \\ 3\sin(2t) \\ 4t \end{bmatrix}.$$

5

[5 marks]

Solution.

first Derivative $\vec{r}\,'(t) = \begin{bmatrix} (2)(-3\sin(2t)) \\ (2)(3\cos(2t)) \\ 4 \end{bmatrix} \Rightarrow = \begin{bmatrix} -6\sin(2t) \\ 6\cos(2t) \\ 4 \end{bmatrix}$ ✓

Second Derivative : $\vec{r}\,''(t) = \begin{bmatrix} 2(-6\cos(2t)) \\ 2(-6\sin(2t)) \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -12\cos(2t) \\ -12\sin(2t) \\ 0 \end{bmatrix}$

□